Supplementary evidence: beta reduction and physical clearance

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This supplement presents the historical optical campaign and the four-station aperture study (last section), alongside the element-wise aperture model used in the article. It distinguishes historical optical statistics from the particle and loss comparisons of the article. Original reviewed assets and campaign records remain frozen; historical claims are interpreted with their original beam and modeling limitations.

1 Original matching and local response

The historical three-start replay reproduces one feasible success, one infeasible success and one budget failure (Table 1). The successful infeasible candidate illustrates why matching and feasibility are separate; the failed candidate is not another low-residual match. Three starts do not establish optimizer reliability, search topology or feasible volume.

Table 1: Historical replay outcomes. The initial start is nominal, not random. J is the scaled residual sum of squares. Feasibility is assessed after search.

Start	J	peak β_x [m]	peak β_y [m]	outcome
Nominal	1.29×10^{-13}	31.437	50.746	feasible success
Random 1	1.21×10^{-12}	25.329	62.556	infeasible success
Random 2	1284.86	69.921	65.043	budget failure

The archived raw/rounded strengths remain feasible at 40-step resolution and independent-map precision. Table 2 supplies numerical values; decimal digits do not prescribe power-supply accuracy. Optical functions, dispersion and projected RMS sizes appear in Fig. 1. Beta peaks reported here are element-boundary samples rather than continuous maxima.

Table 2: Reference and alternative settings [m^{-2}], with reference gradient [$\mathrm{T\,m}^{-1}$] and assumed-bore pole field [T]. Negative K indicates defocusing in x.

Quad	Reference K	Alternative K	Reference G	$\lvert B_{\mathrm{pole}}\rvert$
q11	-0.367525	0.060274	-4.904	0.095
q12	-1.216433	-2.126974	-16.230	0.314
q13	1.586072	1.869755	21.162	0.409
q21	-1.068373	0.954499	-14.255	0.276
q22	2.823108	-0.258649	37.667	0.729
q23	-2.490968	-1.197779	-33.236	0.643
q31	2.042678	2.817163	27.255	0.527
q32	-1.434271	-3.000000	-19.137	0.370
q33	0.351115	0.901619	4.685	0.091

For local sensitivity diagnostics we normalize rows by the optical scales and columns by an assumed common $0.01\,\mathrm{m}^{-2}$ tuning increment: $A_{ij} = (\partial O_i/\partial K_j)\Delta K_j/\sigma_i$. Central steps $1 \times 10^{-4}\,\mathrm{m}^{-2}$, $1 \times 10^{-5}\,\mathrm{m}^{-2}$ and $1 \times 10^{-6}\,\mathrm{m}^{-2}$ converge; the six active modes and three-dimensional local null space have condition number 57.78074. Six finite small-mode perturbations remain inside bounds and agree with linear predictions within 1 %. This differs from the archived mixed-unit condition number 1421.636. Neither Euclidean singular-vector orthogonality nor a local null space guarantees finite optical decoupling. Achieved commissioning correction claims are withdrawn: measurements, actuator calibration and estimation uncertainty have not been established.

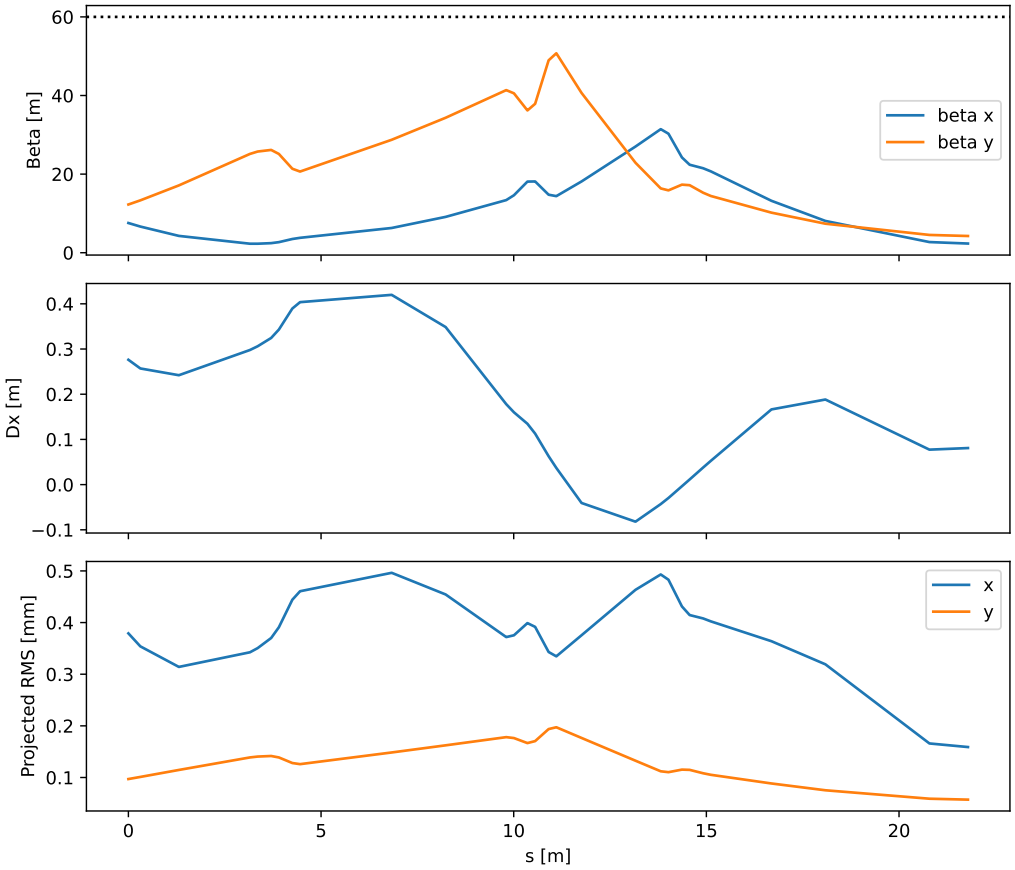


Figure 1: Reference element-boundary beta, horizontal dispersion and projected RMS sizes for the design beam. The dotted beta line is the assumed optical screening ceiling, not a vacuum-chamber edge.

2 Historical optical statistics

Reconstruction of all 550,000 historical observations confirms five optical-rule exceedances, no invalid evaluations, and compatible model/target/error inputs across four runs. The rule is $\beta_{x,\max}$ or $\beta_{y,\max} > 60.01\,\mathrm{m}$, or $\mathcal{M}_x + \mathcal{M}_y > 2.05$, where $\mathcal{M}_u = \frac{1}{2}\mathrm{Tr}(\Sigma_{u,t}^{-1}\Sigma_{u,o}) - 1$ uses unit-emittance transverse matrices. All five failures exceed vertical beta only (60.23 m–61.64 m); they are not physical losses.

The descriptive optical-failure estimate is $5/550000 = 9.09 \times 10^{-6}$ with exact asymmetric 95 % limits $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$; Wilson limits are $[3.8831 \times 10^{-6}, 2.1283 \times 10^{-5}]$ [1, 2]. The pooled horizontal median 0.00208629 has the order-statistic interval [0.00207403, 0.00209890], not a pooled-bootstrap interval. Reconstructed median, 95th and 99th quantile intervals match the archived analysis.

Three 50,000-sample median-prefix changes (20.830, 11.436 and 10.567)$\times 10^{-6}$ miss the 10^{-5} target; the 400,000 follow-up change 2.395×10^{-6} passes. This is median stability, not rare-tail convergence. The follow-up stopping decision is not archived as predeclared; fixed-N intervals do not certify sequential coverage. Historical one-at-a-time sensitivity is mean absolute change of $\mathcal{M}_x + \mathcal{M}_y$, not J. Seed42 ranks beta-only entrance errors (0.00890374), gradients (0.000861122) and energy (0.000106336). Near-zero alignment/roll mismatch does not certify alignment/coupling tolerances. Historical and new-model data remain separate.

3 Finite loss events and numerical validation

The new inference export contains exact block binomial intervals and independent reconstruction of all nine paired clearance intervals. Zero losses among 200 independent three-shot blocks gives one-sided 95 % upper event probability $1 - 0.05^{1/200} = 0.014867$. A two-sided descriptive binomial calculation for zero losses among 1.2 million particles gives upper probability 3.0741×10^{-6}; this requires an independence assumption not justified as unconditional machine reliability. Exact-binomial attribution is Clopper and Pearson [2].

The complete source-labelled covariance, linearization about actual shot centroids, uncensored nonlinear Gaussian quadrature, integration/probe convergence and spatial meshes are available in the attachment. Quadrature certifies moments only on checked cases, not rare-tail loss probabilities. Comparison retains first-loss elements, coupled covariance and exact random seeds. Station transmission ends at m4; unknown continuous chamber and ring capture remain outside the model.

The initial shared-aperture hypothetical run is excluded and retained with exact generating sources. The corrected driver independently detaches the selected station, verified by a regression test. No numerical failure record is silently counted as physical success.

4 Four-station model with hypothetical internal restrictions

This section presents the four-station study, in which apertures are placed only at markers m1–m4. In that model the nominal minimum is fixed by the line's ends (Section 3.1 of the article), so positive results required the hypothetical internal restrictions below. The article replaces this model by element-wise apertures; the numbers here are retained for traceability.

4.1 Frozen strategies and hypothetical restrictions

All settings share entrance/target optics, quadrupole bounds and beam assumptions. Six exit residuals are scaled by $(0.05\,\mathrm{m}, 0.05\,\mathrm{m}, 0.01, 0.01, 0.002\,\mathrm{m}, 0.001)$. Nominal optimization imposes six residual equalities and horizontal peak beta $\leq 60\,\mathrm{m}$. Direct clearance uses a millimeter-scaled epigraph: maximize t subject to $t \leq c_u(s)$ at every station/plane. Two starts (reference and refined low beta) use 200-iteration budgets and 10^{-8} termination tolerance. All six searches converge; their iterations, settings, active bounds and independently checked feasibility are supplied. Exit residuals satisfy declared scales and horizontal screening is checked at 64 subdivisions. Entrance-ceiling ties choose the smallest reference-setting change; their tiny numerical effects are not interpreted as physical gains.

Three predeclared regimes keep entrance/exit limits unchanged: baseline; a horizontal 3 mm halfwidth only at m2 ($s = 6.833\,\mathrm{m}$); and a vertical 1.5 mm halfwidth only at m3 ($s = 11.761\,\mathrm{m}$). Other limits remain unchanged. These narrow internal restrictions are deliberate hypothetical stress cases, not surveyed hardware. A nominal halfwidth sweep of 1.5 mm, 2 mm, 3 mm, 5 mm and 30 mm displays changes in the active restriction without using test outcomes for design selection. The legacy lattice shares an aperture

object at its internal stations; the new driver detaches the selected station before editing its limit. An invalid initial run that inadvertently narrowed both stations is retained separately and excluded.

Settings are frozen before a new independently seeded test. Static errors are independent Gaussian quadrupole relative gradients 0.1 %, x/y offsets 100 µm, rolls 0.5 mrad and dipole relative fields 0.05 %. Each of 200 machines has three independent shots with 5 % entrance beta fluctuations (alpha fixed), horizontal position RMS 0.5 mm, angle RMS 0.2 mrad and centroid momentum RMS 0.1 %. These error levels remain modeling assumptions. Each shot has 2,000 untruncated Gaussian particles with the complete design-beam covariance; all nine regime/strategy combinations receive common errors and particle draws. No optical correction or per-shot retuning is used. The total is 10.8 million particle transports, 1.2 million per combination, not 10.8 million machines.

The primary contrast is alternative minus reference in machine-average minimum five-sigma clearance. Ten thousand paired machine-cluster bootstrap replicates give pointwise descriptive 95 % percentile intervals. No simultaneous coverage claim is made across strategies/regimes. The predeclared useful-effect scale 0.1 mm is illustrative, not a hardware requirement. First loss is nonfinite x after an element; all six coordinates of survivors must be finite. Element exit observations and first-loss indices are retained. Transmission is a separate observable. No additional samples were added after inspecting outcomes.

4.2 Saturation and heterogeneous original outcomes

At zero error, both original settings have minimum clearance 17.455 mm at the entrance (Table 3), the analytic boundary value of Section 3.1 of the article. The full station audit reconstructs all 600 saved paired minima within 1×10^{-12} m. Every one of the 406 exact shot ties has the horizontal entrance active in both designs. The reference has 449 entrance-x, 103 exit-x and 48 exit-y limiting shots; the old low-beta setting has 467, 113 and 20 (Fig. 2). Neither internal station limits that original ensemble.

Table 3: Original nominal station clearances [mm], before spatial reoptimization. R denotes reference and L the old low-beta setting. The entrance caps the full minimum.

s [m]	$c_{x,R}$	$c_{x,L}$	$c_{y,R}$	$c_{y,L}$
0.000	17.455	17.455	18.865	18.865
6.833	27.518	27.416	29.258	29.388
11.761	28.123	28.129	29.117	29.320
21.789	18.555	18.555	19.064	19.064

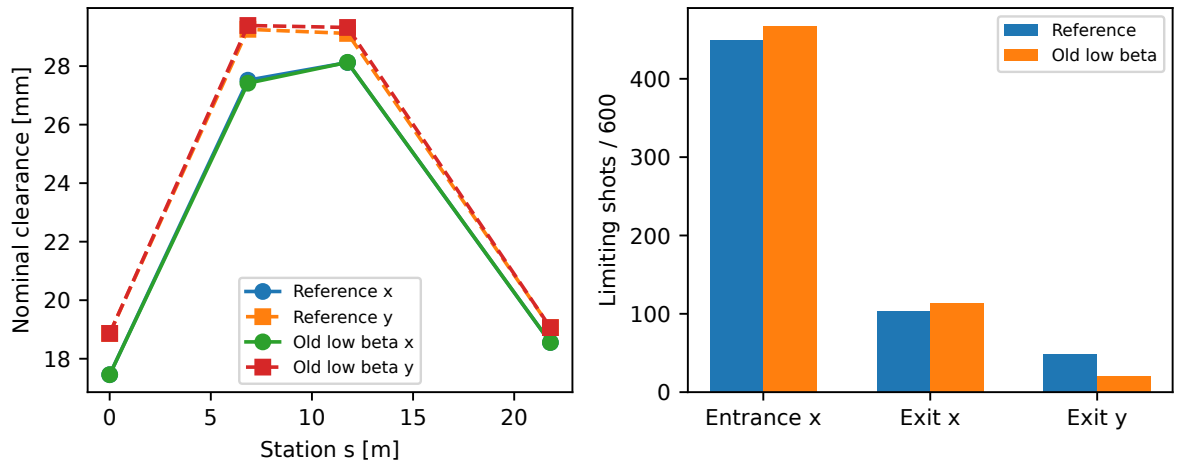


Figure 2: Original affine-model audit. Left: nominal station clearances for both settings and planes. Right: active-limit counts in 600 paired shots, exposing entrance saturation and exit switches. These retrospective data are separate from the new test and refined setting.

The 200 original machine averages include 54 gains, 53 losses and 93 exact ties. Difference standard deviation is 0.236 mm, with 5th/median/95th percentiles −0.322 mm/0 mm/0.403 mm and range −0.923 mm

to 1.401 mm. Its near-zero mean therefore conceals offsetting responses. Excluding the entrance gives a secondary downstream-only mean difference 0.056 mm; this diagnostic does not replace the original full-minimum estimand. Historical optical-rule failures, matching replay and local response diagnostics are retained in the supplement.

4.3 Entrance and exit phase-space distributions

Figure 3 provides a nominal particle-level view of the configured boundary match. A shared sample of 20 000 particles from the full Gaussian entrance covariance is tracked once through each frozen baseline setting at 4 GeV, with 40 integration steps per magnet. No static machine errors or shot jitter are applied. Observations are made before the first element at m1 and after the last element at m4; all four configured aperture stations remain active. A particle is counted as lost if any exit coordinate is nonfinite. All 20 000 particles survive in both settings; this finite nominal demonstration is separate from the error-ensemble loss study and does not establish real chamber transmission or ring capture.

For these plots, AT canonical momenta are converted to geometric slopes, $x' = p_x/p_s$ and $y' = p_y/p_s$, where $p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}$ in units of reference momentum. Horizontal projections include momentum spread and dispersion, rather than representing betatron emittance alone. The tilted entrance clouds rotate and change their projected sizes during transport. The reference and refined low-beta exit clouds are nearly coincident, consistently with their common boundary match; reducing an interior vertical beta peak does not imply a smaller vertical beam at the fixed exit. Tracked transverse RMS values agree with independent linear transport of the same sample to within 0.009 % in this nominal case. The clouds illustrate transport and boundary matching, not a fitted gain in brightness or a new robustness estimate.

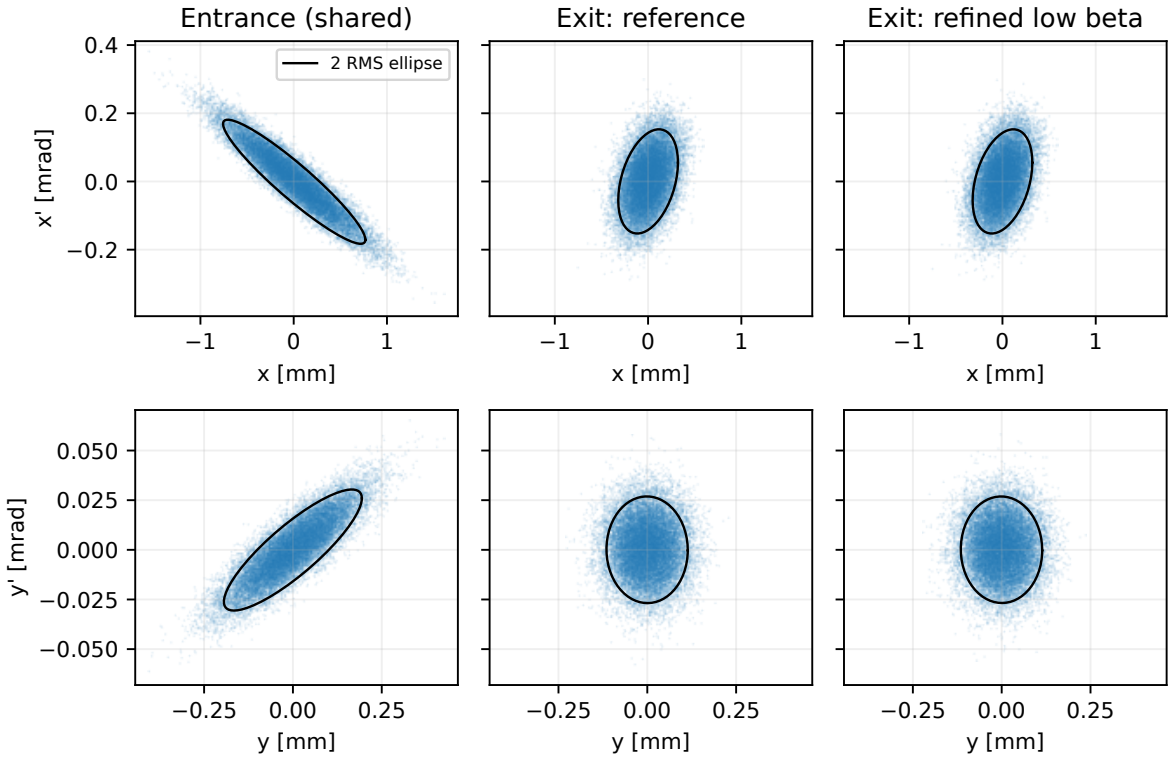


Figure 3: Nominal transverse phase-space distributions: x–x' (top) and y–y' (bottom), at the shared entrance (left), reference exit (center) and refined low-beta exit (right). Positions are in mm and geometric slopes in mrad; axes share scales within each row. Blue points show all 20 000 tracked particles per panel. Black curves, drawn with Accelerator Toolbox's covariance plotting function, are sample-centered two-RMS covariance ellipses; they are not aperture boundaries. No machine errors or shot jitter. Both settings have zero sampled losses at the configured stations.

4.4 Aperture-dependent strategy outcomes on new machines

The spatially refined beta profiles are shown in the main article. The fresh baseline low-beta change is 0.0103 mm with interval [−0.0156 mm, 0.0377 mm], again unresolved. This does not prove exact equivalence or universal inferiority of low-beta design.

Table 4: Frozen-strategy comparisons. Nominal minimum and new paired mean change are in mm; brackets give pointwise 95 % machine-cluster intervals. Restricted cases are hypothetical. The baseline clearance-setting change is numerical.

Regime	Strategy	Nominal	Mean change	95% interval
Baseline	low beta	17.455	+0.010	[-0.016, +0.038]
Baseline	clearance	17.455	–	degenerate
x restriction	low beta	0.420	-0.152	[-0.157, -0.147]
x restriction	clearance	1.142	+0.881	[+0.847, +0.914]
y restriction	low beta	0.821	+0.073	[+0.048, +0.098]
y restriction	clearance	0.915	+0.212	[+0.174, +0.250]

At the horizontal restriction, low beta worsens the mean by 0.152 mm, while direct nominal clearance gains 0.881 mm (Table 4). Nominal horizontal betatron RMS changes from 0.220 mm to 0.230 mm in the direct design, but dispersive RMS decreases from 0.445 mm to 0.292 mm. On the held-out shots, mean absolute horizontal centroid decreases from 1.218 mm to 0.960 mm and mean five-RMS size decreases from 2.485 mm to 1.862 mm. Their combined changes explain the gain; minimizing vertical beta alone does not control either term.

At the vertical restriction, low beta gains 0.073 mm and direct clearance gains 0.212 mm. Low beta reduces mean five-RMS size from 0.885 mm to 0.681 mm, but increases mean absolute centroid from 0.230 mm to 0.360 mm; much of its nominal size benefit is offset by trajectory sensitivity. Direct clearance gives 0.586 mm and 0.316 mm for these terms. The low-beta interval excludes zero but lies below the illustrative 0.1 mm useful-effect scale; direct gains exceed that scale in both restricted cases. These are outcomes for the selected finite searches and assumptions, not universal strategy dominance.

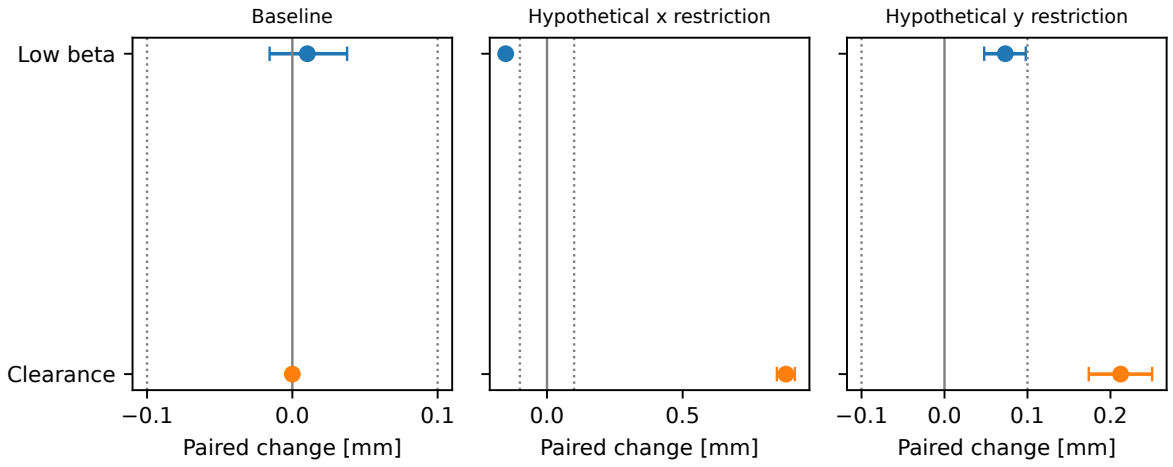


Figure 4: New paired mean clearance changes with pointwise 95 % intervals for the 200 independent machines. Dotted vertical lines indicate the declared illustrative ±0.1 mm scale. Every regime includes the entrance and exit in its minimum.

4.5 Physical particle loss is a separate outcome

No baseline losses occur. For zero of 200 independent blocks, the one-sided 95 % upper probability of an observed-loss block is 1.4867 % [2]. The event consists of three sampled shots with 2,000 particles per shot; it is not the probability of any loss during indefinite operation. Shared machine and shot randomness

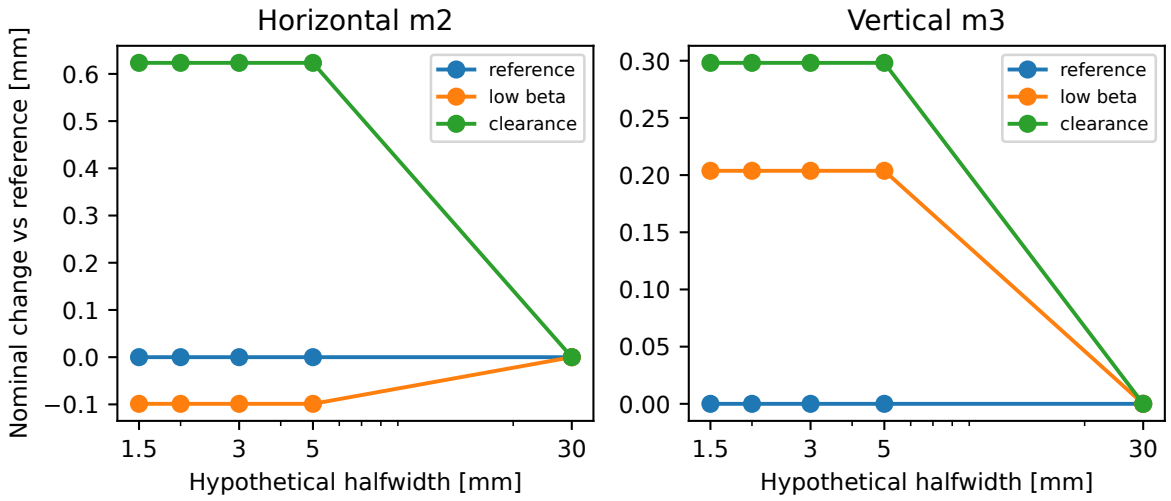


Figure 5: Nominal clearance changes versus reference across halfwidth sweeps, with frozen settings. The adjusted plane is horizontal m2 or vertical m3; all other limits stay fixed. A wide restriction returns to the entrance ceiling. Hypothetical sweeps expose metric saturation; they are not a prediction of chamber dimensions.

Table 5: Observed first losses in the new test: 1.2 million particles and 200 three-shot machine blocks per row. A block is affected if any sampled particle is lost. Counts refer to model stations, not continuous-chamber or ring losses.

Regime	Strategy	Lost particles	Affected blocks
Baseline	reference	0	0
Baseline	low beta	0	0
Baseline	clearance	0	0
x restriction	reference	65798	155
x restriction	low beta	79736	163
x restriction	clearance	17200	72
y restriction	reference	1	1
y restriction	low beta	43	6
y restriction	clearance	26	2

prevents interpreting all particles as independent machine reliability trials. The supplement gives exact block intervals and separately labelled descriptive particle bounds.

The vertical restriction illustrates that a higher mean clearance need not produce fewer losses: reference/low-beta/direct settings lose 1/43/26 particles in 1/6/2 blocks (Table 5). These few tail events do not establish a converged machine reliability ranking. In the stronger horizontal stress case, loss counts and affected blocks are lower for direct clearance than reference, and higher for low beta. Paired transmission intervals are supplied separately; clearance intervals must not be reused as loss-probability statements.

References

[1] E. B. Wilson. Probable inference, the law of succession, and statistical inference. *J. Am. Stat. Assoc.*, 22:209–212, 1927. `https://doi.org/10.1080/01621459.1927.10502953`.

[2] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. `https://doi.org/10.1093/biomet/26.4.404`.